Data Collection and Operationalization

September 12, 2022

Outline

1. Coding Human Right Violations
2. Collecting Data/Sampling
3. Levels of Measurement
4. Codebooks and Datasets
5. R: read in, view, tabulate, and extracting variables

Coding Human Rights Violations

1. Countries under a secure rule of law, people are not imprisoned for their view, and torture is rare or exceptional. Political murders are extremely rare.
2. There is a limited amount of imprisonment for nonviolent political activity. However, few persons are affected, torture and beatings are exceptional. Political murder is rare.
3. There is extensive political imprisonment, or a recent history of such imprisonment. Execution or other political murders and brutality may be common. Unlimited detention, with or without a trial, for political views is accepted.
4. Civil and political rights violations have expanded to large numbers of the population. Murders, disappearances, and torture are a common part of life. In spite of its generality, on this level terror affects those who interest themselves in politics or ideas.
5. Terror has expanded to the whole population. The leaders of these societies place no limits on the means or thoroughness with which they pursue personal or ideological goals.

Coding Human Rights Violations

1. Did these coding rules produce valid measures of human rights violations?

2. Did these coding rules produce reliable measures of human rights violations?

Initial Terminology

- Variable- A thing that varies. Has a *label/name* and *values*.
- Data (plural)- The values for a variable.
- Unit of Analysis- The thing you are studying; *unique* categorization of observations being analyzed (think of rows in a spreadsheet)
  - e.g., individual, state, country, dyad, country-year, etc.
  - Should be reflective of your research question
  - Data for all variables should be at or above the same level

Where to get data?

- Large dataset projects
- Replication data from published research
- Archives or secondary resources
- Surveys

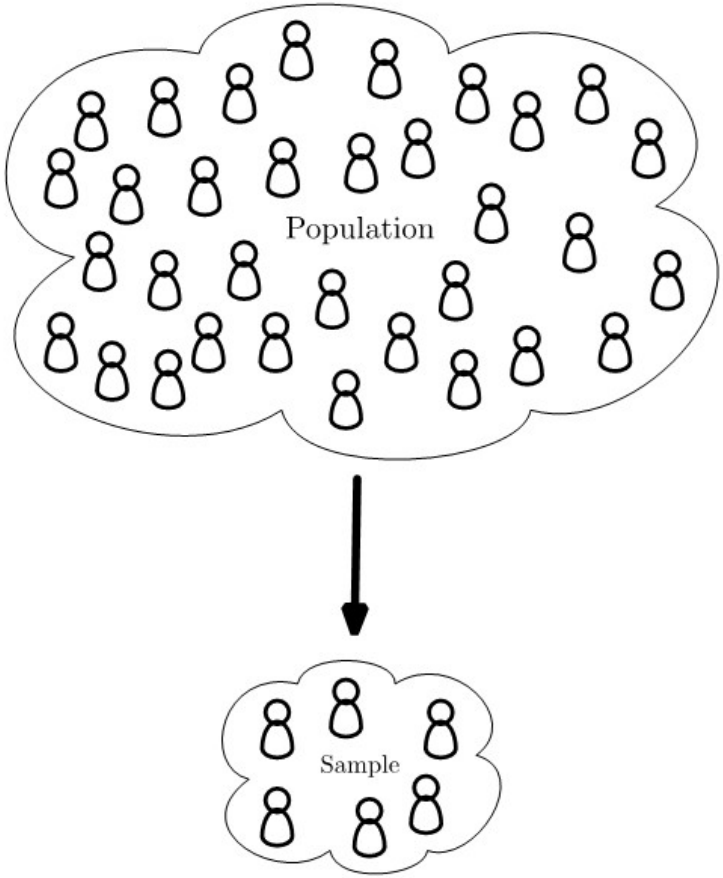
Collecting Data: Sampling

Population: The group you want to make conclusions about

Sample: The data you have (often a subset of the population)

Goal of sampling: *generalize* to the population without explicitly observing the whole population

- Ideal: Random Sample- each case has an equal chance of being selected
- If not: Make sure selection rule not related to DV

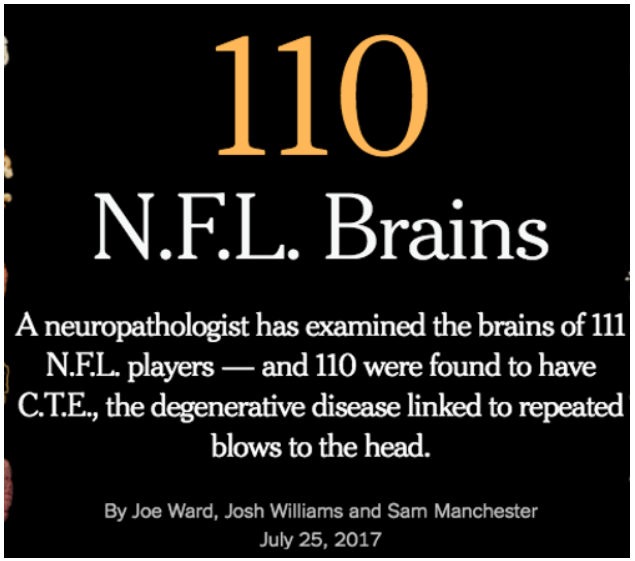


Selection Bias

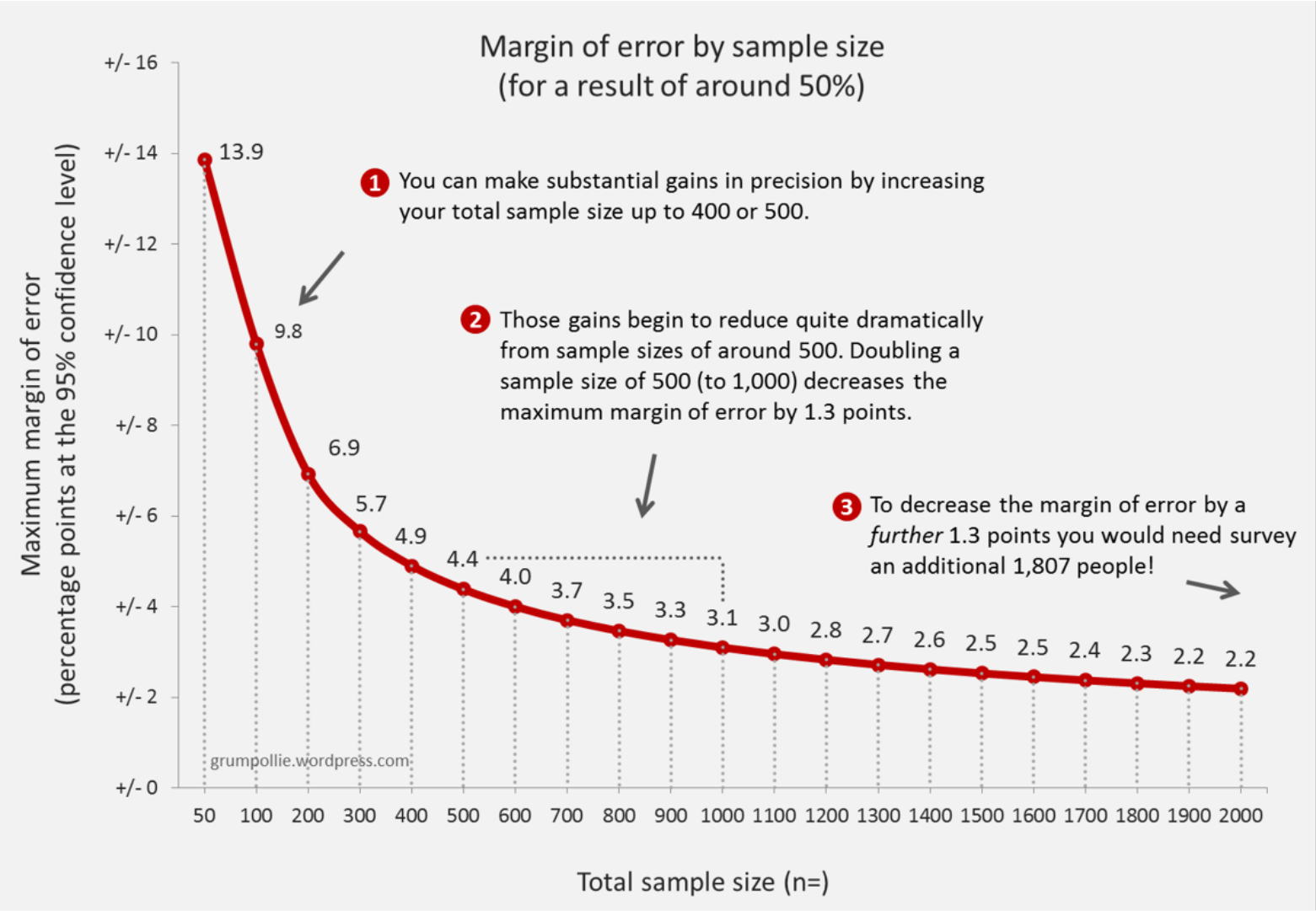
When a factor related to the dependent variable is related to the observations/subjects selected for a sample.

Examples:

- Chronic traumatic encephalopathy (CTE) in NFL brains
- WWII British RAF plane armor



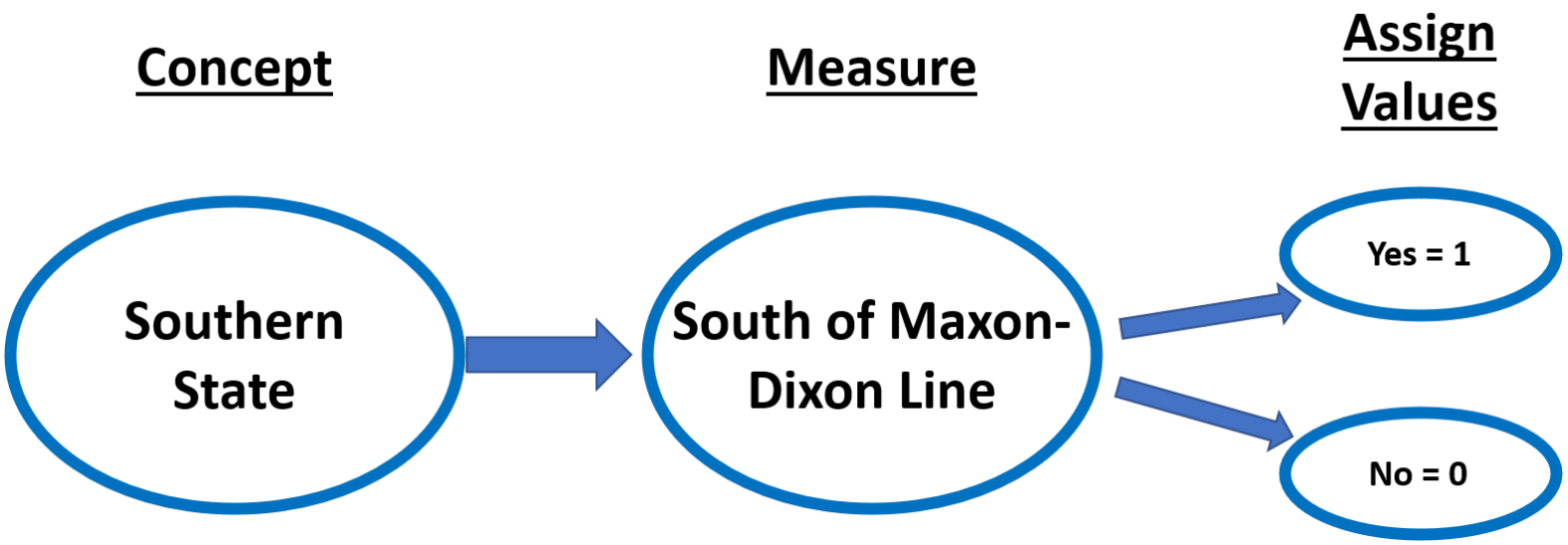
Collecting Data: How Much Do I Need?



Operationalization and Levels of Measurement

Measurement- Defining observable units of abstract concepts (defining "coding rules")

Operationalization- Assigning numerical values to variables



Levels of Measurement: Categorical Variables

(a.k.a. "Nominal" variables) are those whose values equate or differentiate observations but have no ordered ranking.

e.g., party id, brands of tuna, biological taxonomy

Levels of Measurement: Ordinal Variables

those whose values can be rank ordered but do not have equal unit differences.

<u>Likert Scale</u>
1 = "Strongly Agree" 2 = "Agree" 3 = "Neutral" 4 = "Disagree"
5 = "Strongly Disagree"

Levels of Measurement: Continuous Variables

(a.k.a. "Interval/Ratio" variables) are those whose values can be rank ordered and have equal unit differences.

e.g., calendar year, income, age, percentages

Designing Coding Schemes

- Values of variable should be exhaustive
- Establish that the values are mutually exclusive
- Try to create more (rather than fewer) values
- Document your coding scheme in a codebook

Codebooks

WATERACCESS
Interval

Value = percent of the population that has regular access to clean drinking water
. = missing data

Source: bibliographic entry for the source of information

Codebooks

WATERACCESS
Ordinal

0 = 0-10% of the population has access to clean drinking water
1 = 11-20% of the population has access to clean drinking water
2 = 21-30% of the population has access to clean drinking water
3 = 31-40% of the population has access to clean drinking water
4 = 41-50% of the population has access to clean drinking water
5 = 51-60% of the population has access to clean drinking water
6 = 61-70% of the population has access to clean drinking water
7 = 71-80% of the population has access to clean drinking water
8 = 81-90% of the population has access to clean drinking water
9 = 91-100% of the population has access to clean drinking water
. = missing data

Source: bibliographic entry for the source of information

Dataset Example

COW Datasets Link

To R!